Run a Paired Spatial MEM Cross-Validation Fixture

Description

Runs exact and fast spatial MEM construction through identical fold preparation, sjSDM fitting, and held-out prediction work.

Usage

```
run_spatial_mev_paired_cv_fixture(
  data_community_matrix = NULL,
  data_abiotic_wide = NULL,
  data_coords_projected = NULL,
  data_sample_ids = NULL,
  data_assignments = NULL,
  candidate = NULL,
  sel_abiotic_formula = NULL,
  config_model_fitting = NULL,
  config_data_processing = NULL,
  device = "cpu",
  prepare_function = prepare_sjsdm_cross_validation_fold,
  fit_function = fit_sjsdm_regularization_candidate,
  predict_function = predict_sjsdm_probability_matrix
)
```

Arguments

`data_community_matrix`, `data_abiotic_wide`, `data_coords_projected`	Model inputs accepted by [prepare_sjsdm_cross_validation_fold()].
`data_sample_ids`	Sample metadata accepted by [prepare_sjsdm_cross_validation_fold()].

`data_assignments`	Table containing 'repeat_id', 'fold_id', and 'row_index'. Each row index must occur once in every repeat.
`candidate`	One regularization candidate accepted by [fit_sjsdm_regularization_candidate()].
`sel_abiotic_formula`	Abiotic formula passed to [fit_sjsdm_regularization_candidate()].
`config_model_fitting`, `config_data_processing`	Shared model and data-processing configurations.
`device`	Device passed to [fit_sjsdm_regularization_candidate()].
`prepare_function`, `fit_function`, `predict_function`	Injectable production adapters used for focused orchestration tests.

Value

Named list containing paired benchmark rows, predictions, and fold diagnostics.